Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

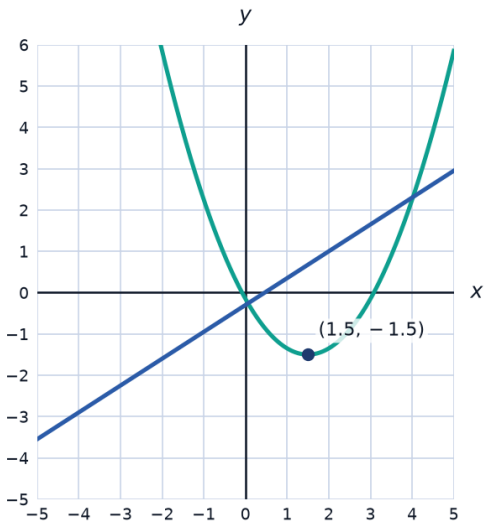
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Basic probability**

Two fair six-sided dice are rolled. Find the probability their sum is 8.

2 **[Foundational] • Target: Permutations and combinations**

How many ways can a committee of 4 be chosen from 10 people?



3 **[Foundational] • Target: Permutations and combinations**

How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?

4 **[Foundational] • Target: Permutations and combinations**

Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.

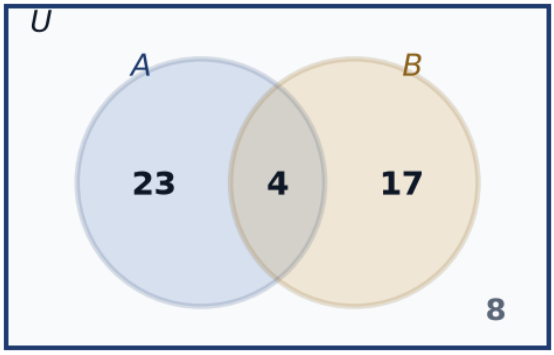
5 **[Foundational] • Target: Independent events and the addition rule**

Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

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6 [Foundational] • Target: Complement rule

Events A and B satisfy $P(A) = 0.60$, $P(B) = 0.50$, and conditional probability $P(A|B) = 0.40$. Determine $P(A \cup B)$ and state whether A and B are independent.



7 [Foundational] • Target: Permutations and combinations

A quality control inspector tests components that each have an independent 10% defect rate ($p = 0.10$). If a random batch of 5 components is tested: (a) Find the probability that exactly 2 components are defective. (b) Find the probability that at least 1 component is defective.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Independent events and the addition rule

Two events A and B are independent. $P(A) = 0.4$, $P(B) = 0.5$. Find $P(A \cap B)$.

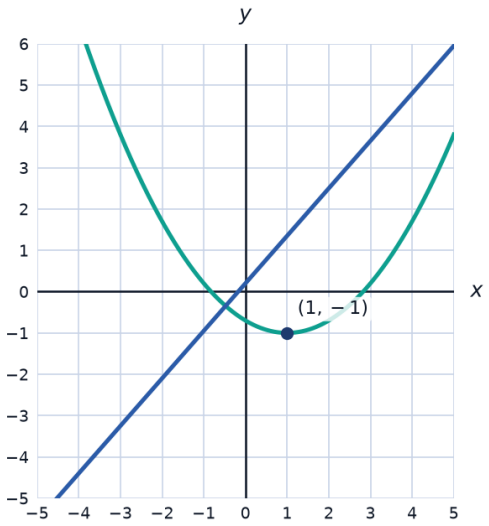
9 [Intermediate] • Target: Basic probability

A delegation of 5 delegates is chosen from 6 men and 8 women. (a) In how many ways can the delegation be selected if it must contain exactly 3 women? (b) What is the exact probability that the delegation contains at least 4 women?

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10 [Intermediate] • Target: Complement rule

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.



11 [Intermediate] • Target: Permutations and combinations

A dataset of 40 exam scores has mean $\bar{x} = 72$ and sample standard deviation $s = 8$. The instructor rescales all scores using $y_i = 1.25x_i - 10$. What are the mean and standard deviation of the rescaled scores?

12 [Intermediate] • Target: Independent events and the addition rule

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

13 [Intermediate] • Target: Basic probability

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

14 [Intermediate] • Target: Complement rule

Compute $(\sqrt{3} - i)^8$ using De Moivre's Theorem, expressing the result in standard rectangular form $a + bi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

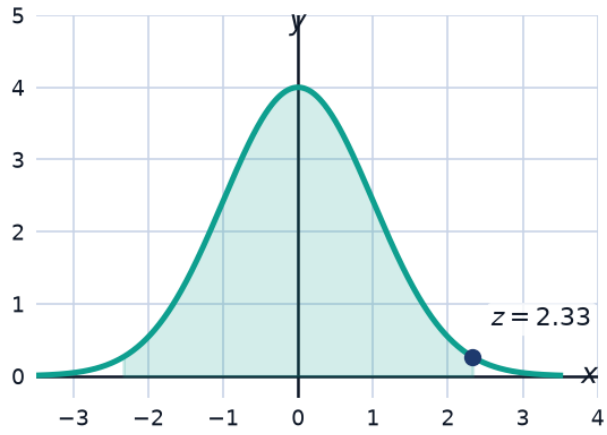
15 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 5.01 Introductory Calculus

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

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16 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 1.01 Complex Numbers

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.



17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Sequences & Series

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ \ldots\ + (2n - 1) = n^2$.

18 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

19 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

In triangle ABC, $a = 10$, $b = 16$, and $A = 30^\circ$. Determine whether $0, 1,$ or 2 distinct triangles can be formed, and calculate all possible values for angle B to the nearest tenth of a degree.

20 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

In triangle ABC, $a = 7$, $b = 5$, and $\cos C = \frac{1}{5}$. (a) Find the exact length of side c. (b) Calculate the exact value of $\sin^2 C$ and the area of triangle ABC.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic probability

A bag contains 9 yellow marbles, 5 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{1}{3}$

b. $\frac{3}{7}$

c. $\frac{10}{21}$

d. $\frac{5}{21}$

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22 [Foundational] • Target: Complement rule

A bag contains 3 green marbles, 7 yellow marbles, and 7 red marbles. A marble is drawn at random. Find $P(\text{green})$.

a. $\frac{7}{17}$

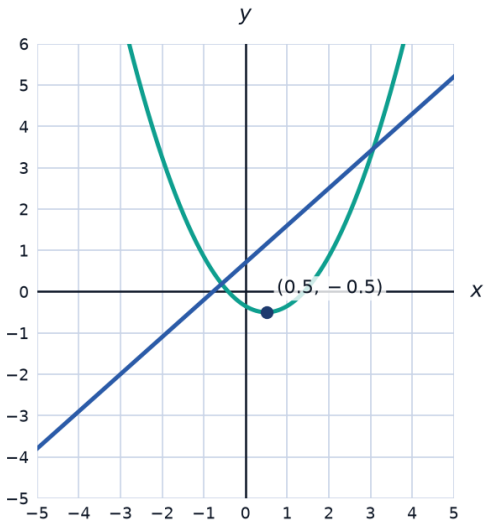
b. $\frac{5}{17}$

c. $\frac{4}{17}$

d. $\frac{3}{17}$

23 [Foundational] • Target: Permutations and combinations

A bag contains 7 blue marbles, 7 green marbles, and 3 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.



a. $\frac{9}{17}$

b. $\frac{8}{17}$

c. $\frac{3}{17}$

d. $\frac{7}{17}$

24 [Foundational] • Target: Independent events and the addition rule

A bag contains 3 red marbles, 3 blue marbles, and 4 green marbles. A marble is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{10}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$

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25 [Foundational] • Target: Basic probability

How many ways can 2 distinct positions be filled from 12 candidates, where order matters?

a. 66

b. 138

c. 132

d. 126

26 [Foundational] • Target: Complement rule

Out of 38 equally likely tiles, 5 are red, 8 are numbered, and 3 are both. One item is chosen at random. Find $P(\text{red or numbered})$.

a. $\frac{5}{19}$

b. $\frac{6}{19}$

c. $\frac{13}{38}$

d. $\frac{4}{19}$

27 [Foundational] • Target: Permutations and combinations

Out of 27 equally likely boxes, 7 are labeled, 8 are sealed, and 7 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.

a. $\frac{2}{9}$

b. $\frac{10}{27}$

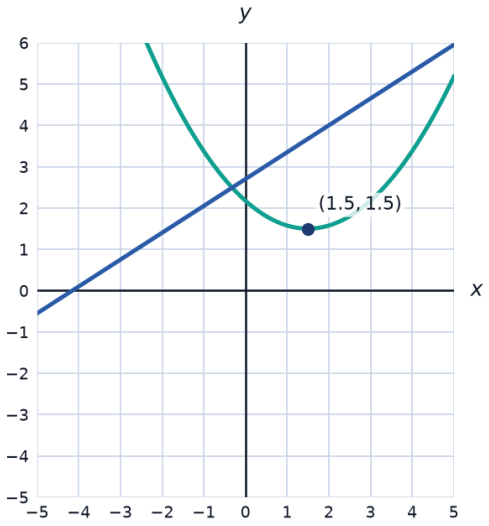
c. $\frac{8}{27}$

d. $\frac{5}{9}$

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28 [Foundational] • Target: Independent events and the addition rule

Out of 26 equally likely tickets, 9 are gold, 12 are striped, and 8 are both. One item is chosen at random. Find $P(\text{gold or striped})$.



a. $\frac{21}{26}$

b. $\frac{15}{26}$

c. $\frac{1}{2}$

d. $\frac{11}{26}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Basic probability

Out of 37 equally likely numbers, 4 are even, 10 are prime, and 2 are both. One item is chosen at random. Find $P(\text{even or prime})$.

a. $\frac{14}{37}$

b. $\frac{10}{37}$

c. $\frac{12}{37}$

d. $\frac{15}{37}$

30 [Intermediate] • Target: Complement rule

Events A and B are mutually exclusive, with $P(A) = 2/7$ and $P(B) = 3/8$. Find $P(A \text{ or } B)$.

a. 37/56

b. 269/280

c. 55/168

d. 83/168

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31 **[Intermediate]** • **Target: Permutations and combinations**

A survey was conducted on the preferred modes of transportation for students travelling to school. The results are summarized in the two-way table below. If a student is chosen at random from those surveyed, and it is known that they prefer to travel by bus, what is the probability that they are in Year 10? Let N_{10} represent the number of Year 10 students who prefer the bus, and N_{Bus} represent the total number of students who prefer the bus. The probability we are looking for is $\frac{N_{10}}{N_{Bus}}$. | Year Group | Bicycle | Bus | Train | Walk | Total | |---|---|---|---|---|---| | Year 9 | 45 | 60 | 25 | 70 | 200 | | Year 10 | 55 | x | 35 | 80 | 270 | | Year 11 | 30 | 50 | 40 | 50 | 170 | | Total | 130 | 160 | 100 | 200 | 640 |

a. $\frac{130}{160}$

b. $\frac{x}{160}$

c. $\frac{x}{270}$

d. $\frac{160}{640}$

32 **[Intermediate]** • **Target: Independent events and the addition rule**

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD.

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

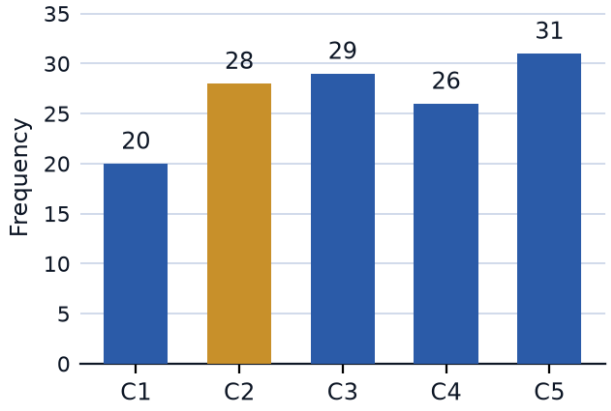
c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

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33 [Intermediate] • Target: Basic probability

The bar chart below shows the number of students who chose different sports as their favorite. What is the total number of students surveyed?



a. 50

b. 60

c. 70

d. 80

34 [Intermediate] • Target: Complement rule

The following bar chart shows the number of students who prefer different fruits. What is the total number of students who prefer apples and bananas?

a. 20

b. 25

c. 30

d. 35

35 [Intermediate] • Target: Permutations and combinations

The bar chart below shows the number of students who achieved different scores on a recent quiz. What is the total number of students who scored 8 or 10?

a. 25

b. 30

c. 35

d. 40

36 [Intermediate] • Target: Independent events and the addition rule

The table shows exam marks for 20 students. Find the mean and identify the modal class.

a. Mean ≈ 65, Modal class 60–69

b. Mean ≈ 68, Modal class 70–79

c. Mean ≈ 60, Modal class 50–59

d. Mean ≈ 72, Modal class 60–69

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37 [Intermediate] • Target: Basic probability

In a group of 20 students, 9 study French, and 1 study both French and Spanish. If a student is chosen at random from those studying French, what is the probability they also study Spanish?

a. 1/9

b. 7/36

c. 19/90

d. 5/18

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

A bag has 23 marbles, of which 2 are red. What is the probability of drawing a red marble?

a. 43/230

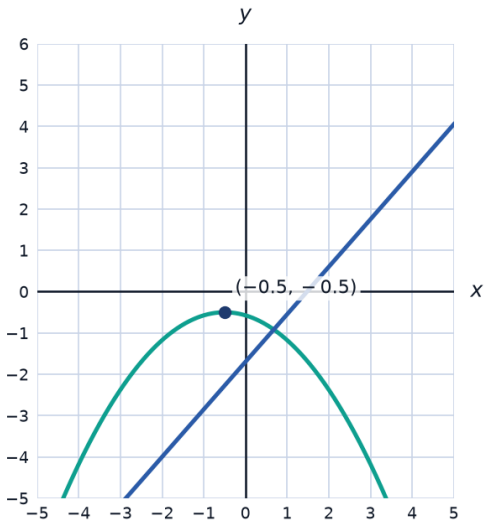
b. 2/23

c. 89/230

d. 33/115

39 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

In how many ways can 2 of 7 books on a shelf be arranged?



a. 82

b. 33

c. 42

d. 341

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40 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

In how many ways can 3 of 4 photos in an album be arranged?

a. 132

b. 69

c. 154

d. 24

41 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Complex Numbers

In how many ways can 2 of 4 runners in a race (1st through r-th place) be arranged?

a. 296

b. 112

c. 12

d. 270

42 [Challenging] • Target: Complement rule • Cross-Lesson: 2.01 Sequences & Series

Find the population variance of the data set: $2, 8, 12, 8, 10$.

a. 173/15

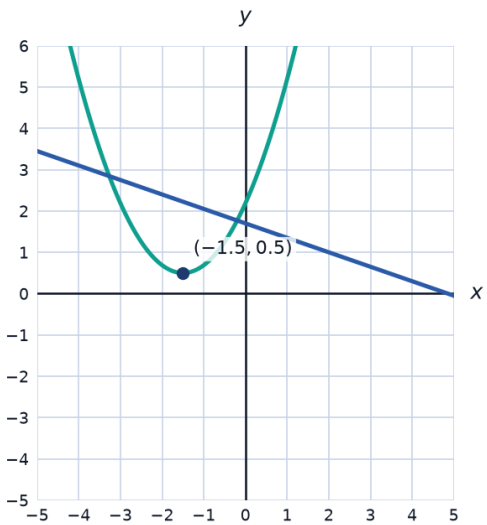
b. 56/5

c. 677/60

d. 163/15

43 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 3.01 Functions & Graphs

A basketball player makes a free throw with probability $2/3$. If she takes 2 free throws, what is the probability she makes at least one?



a. 53/90

b. 55/72

c. 8/9

d. 35/36

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44 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 4.01 Matrices & Vectors

A committee of 5 people is chosen from 9 people, and then 4 of those 5 are assigned distinct roles (e.g., chair, secretary, ...). How many ways can this be done?

a. 2696

b. 15120

c. 396

d. 4484

45 [Challenging] • Target: Basic probability • Cross-Lesson: 5.01 Introductory Calculus

Find the median of the data set: $9, 4, 9, 2, 7$.

a. 17

b. 24

c. 13

d. 7

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